MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – D)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. The equation of a line parallel to the y-axis and at a distance of 5 units to the right of the y-axis is: [1]

(A) $x = 5$ **(B)** $x = -5$ **(C)** $y = 5$ **(D)** $y = -5$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $x = 5$

Concept: Any line parallel to the y-axis has a constant x-coordinate, i.e., $x = a$.

Formula Used: Line parallel to y-axis at distance a to its right: $x = a$ $(a > 0)$

Since the line is 5 units to the **right** of the y-axis, every point on it has abscissa $x = +5$.

$$\boxed{x = 5}$$

Q2. The equation of a line parallel to the x-axis and at a distance of 3 units below the x-axis is: [1]

(A) $x = 3$ **(B)** $x = -3$ **(C)** $y = -3$ **(D)** $y = 3$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $y = -3$

Concept: Any line parallel to the x-axis has a constant y-coordinate, i.e., $y = b$.

Formula Used: Line parallel to x-axis at distance $|b|$ below it: $y = -b$ $(b > 0)$

Since the line lies 3 units **below** the x-axis, its ordinate is negative: $y = -3$.

$$\boxed{y = -3}$$

Q3. The line represented by the equation $x - y = 0$ is: [1]

(A) parallel to x-axis
(B) parallel to y-axis
(C) passing through the origin
(D) passing through the point $(3, 2)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) passing through the origin

Given: Linear equation $x - y = 0 \implies y = x$.

Formula Used: A line passes through the origin $(0, 0)$ if $(0, 0)$ satisfies its equation.

Substituting $x = 0$ and $y = 0$: LHS $= 0 - 0 = 0 =$ RHS.
Notice it is not horizontal, not vertical, and for $(3, 2)$, $3 - 2 = 1 \neq 0$. Hence it passes through the origin: passing through the origin.

Q4. The distance between the points $(4\cos\theta + 3\sin\theta,\ 0)$ and $(0,\ 4\sin\theta - 3\cos\theta)$ is: [1]

(A) 25 **(B)** 7 **(C)** 5 **(D)** $\sqrt{7}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 5

Given: Points $A(4\cos\theta + 3\sin\theta,\ 0)$ and $B(0,\ 4\sin\theta - 3\cos\theta)$.

Formula Used: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Substituting the coordinates and using $\sin^2\theta + \cos^2\theta = 1$:

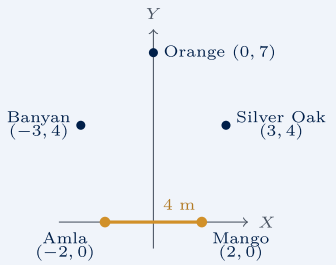
$$\begin{aligned} AB &= \sqrt{(0 - (4\cos\theta + 3\sin\theta))^2 + (4\sin\theta - 3\cos\theta - 0)^2} \\ &= \sqrt{(16\cos^2\theta + 24\sin\theta\cos\theta + 9\sin^2\theta) + (16\sin^2\theta - 24\sin\theta\cos\theta + 9\cos^2\theta)} \\ &= \sqrt{16(\cos^2\theta + \sin^2\theta) + 9(\sin^2\theta + \cos^2\theta)} = \sqrt{16 + 9} = \sqrt{25} = \boxed{\mathbf{5}\text{ units}} \end{aligned}$$

Q5. 5 trees Banyan, Silver Oak, Orange, Amla and Mango were planted in the garden as shown in the figure. If 1 unit = 1 m in the figure, which of the following trees are at a distance of 4 metres? *(From diagram: Orange $(0, 7)$, Silver Oak $(3, 4)$, Banyan $(-3, 4)$, Mango $(2, 0)$, Amla $(-2, 0)$)* [1]

(A) Amla and Mango
(B) Mango and Silver Oak
(C) Silver Oak and Orange
(D) Orange and Mango

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Amla and Mango



Given: Coordinates from grid (1 unit = 1 m): Amla $(-2, 0)$, Mango $(2, 0)$, Silver Oak $(3, 4)$, Banyan $(-3, 4)$, Orange $(0, 7)$.

We apply the distance formula:

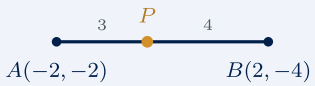
Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Testing distances for each option:
• Amla $(-2, 0)$ to Mango $(2, 0)$: $d = \sqrt{(2 - (-2))^2 + (0 - 0)^2} = \sqrt{4^2} = \mathbf{4}$ m.
• Mango to Silver Oak: $\sqrt{(3 - 2)^2 + (4 - 0)^2} = \sqrt{17}$ m; Silver Oak to Orange: $\sqrt{9 + 9} = 3\sqrt{2}$ m; Orange to Mango: $\sqrt{4 + 49} = \sqrt{53}$ m.

Required pair = Amla and Mango

Q6. The coordinates of the end points of the line segment AB are $A(-2,-2)$ and $B(2,-4)$. P is the point on AB such that $BP = \frac{4}{7}AB$. Find the coordinates of point P. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Finding the Internal Ratio $AP : PB$ [1 Mark]

Given: $A(-2,-2)$, $B(2,-4)$, and $BP = \frac{4}{7}AB$. Then $AP = AB - BP = AB - \frac{4}{7}AB = \frac{3}{7}AB$, so $AP : PB = 3 : 4$.

We know the internal section formula:

Formula Used: $P(x,y) = \left(\frac{m_1x_2 + m_2x_1}{m_1+m_2}, \frac{m_1y_2 + m_2y_1}{m_1+m_2}\right)$

Step 2: Applying the Section Formula [1 Mark]

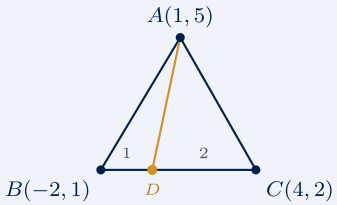
Substituting $m_1 = 3$, $m_2 = 4$, $(x_1, y_1) = (-2,-2)$, and $(x_2, y_2) = (2,-4)$:

$$x = \frac{3(2)+4(-2)}{3+4} = \frac{6-8}{7} = -\frac{2}{7}, \qquad y = \frac{3(-4)+4(-2)}{3+4} = \frac{-12-8}{7} = -\frac{20}{7}$$

$$\boxed{P(x,y) = \left(-\frac{\mathbf{2}}{\mathbf{7}}, -\frac{\mathbf{20}}{\mathbf{7}}\right)}$$

Q7. In the given figure, point D divides the side BC of $\triangle ABC$ in the ratio $1:2$. Find length AD. *(From given figure: $A(1,5)$, $B(-2,1)$, $C(4,2)$)* [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Finding Coordinates of Point D [1 Mark]

Given: $A(1,5)$, $B(-2,1)$, $C(4,2)$, and D divides BC internally in ratio $1:2$.

We apply the internal section and distance formulas:

Formula Used: $D = \left(\frac{m_1x_2 + m_2x_1}{m_1+m_2}, \frac{m_1y_2 + m_2y_1}{m_1+m_2}\right)$

$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

Substituting $m_1 = 1$, $m_2 = 2$, $B(-2,1)$, and $C(4,2)$:

$$D = \left(\frac{1(4)+2(-2)}{1+2}, \frac{1(2)+2(1)}{1+2}\right) = \left(\frac{4-4}{3}, \frac{2+2}{3}\right) = \left(\mathbf{0}, \frac{\mathbf{4}}{\mathbf{3}}\right)$$

Step 2: Calculating Length AD [1 Mark]

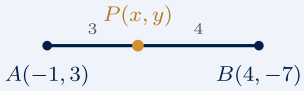
Using the distance formula between $A(1,5)$ and $D\left(0, \frac{4}{3}\right)$:

$$AD = \sqrt{(0-1)^2 + \left(\frac{4}{3}-5\right)^2} = \sqrt{(-1)^2 + \left(-\frac{11}{3}\right)^2}$$

$$= \sqrt{1 + \frac{121}{9}} = \sqrt{\frac{9+121}{9}} = \sqrt{\frac{130}{9}} = \boxed{\frac{\sqrt{\mathbf{130}}}{\mathbf{3}} \text{ units}}$$

Q8. Find the coordinates of the point which divides the line segment joining $(-1, 3)$ and $(4, -7)$ internally in the ratio $3 : 4$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Formula Declaration [0.5 Mark]
Given: $A(-1, 3)$, $B(4, -7)$, and ratio $m_1 : m_2 = 3 : 4$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

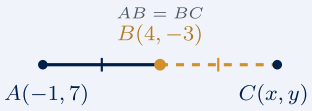
Step 2: Substitution and Evaluation [1.5 Marks]

$$x = \frac{3(4) + 4(-1)}{3 + 4} = \frac{12 - 4}{7} = \frac{8}{7}, \qquad y = \frac{3(-7) + 4(3)}{3 + 4} = \frac{-21 + 12}{7} = -\frac{9}{7}$$

$$\boxed{\text{Required point } = \left(\mathbf{\frac{8}{7}}, \mathbf{-\frac{9}{7}} \right)}$$

Q9. Find the coordinates of the point C which lies on the line AB produced such that $AC = 2BC$, where coordinates of points A and B are $(-1, 7)$ and $(4, -3)$ respectively. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Interpreting C on Line AB Produced [1 Mark]
Given: $A(-1, 7)$, $B(4, -3)$, and $C(x, y)$ lies on AB produced beyond B such that $AC = 2BC$. Since $AC = AB + BC$, we have $AB = BC$, so B is the midpoint of AC.

We apply the midpoint formula:

Formula Used: $B = \left(\dfrac{x_A + x_C}{2}, \dfrac{y_A + y_C}{2} \right)$

Step 2: Finding Coordinates of C [1 Mark]
Equating midpoint of AC to $B(4, -3)$:

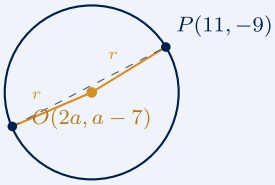
$$\left(\frac{-1 + x}{2}, \frac{7 + y}{2} \right) = (4, -3)$$

$$\frac{-1 + x}{2} = 4 \implies -1 + x = 8 \implies x = 9$$
$$\frac{7 + y}{2} = -3 \implies 7 + y = -6 \implies y = -13$$

$$\boxed{\text{Coordinates of } C = (\mathbf{9}, \mathbf{-13})}$$

Q10. The coordinates of the centre of a circle are $(2a, a-7)$. Find the value(s) of 'a' if the circle passes through the point $(11, -9)$ and has diameter $10\sqrt{2}$ units. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Setting Radius Equality [0.5 Mark]
Given: Centre $O(2a, a-7)$, boundary point $P(11, -9)$, diameter $D = 10\sqrt{2}$ units $\implies$ radius $r = \dfrac{10\sqrt{2}}{2} = 5\sqrt{2}$ units.

We equate radius OP to r:

Formula Used: $OP = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = r$

Step 2: Solving the Quadratic Equation for a [1.5 Marks]
Using $OP^2 = r^2$:

$$(2a - 11)^2 + [(a - 7) - (-9)]^2 = (5\sqrt{2})^2$$

$$(2a - 11)^2 + (a + 2)^2 = 50$$

$$(4a^2 - 44a + 121) + (a^2 + 4a + 4) = 50 \implies 5a^2 - 40a + 125 = 50$$

$$5a^2 - 40a + 75 = 0 \implies a^2 - 8a + 15 = 0 \quad \text{(dividing by 5)}$$

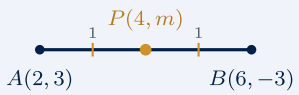
Factoring by splitting the middle term:

$$a^2 - 5a - 3a + 15 = 0 \implies (a-5)(a-3) = 0 \implies \boxed{\mathbf{a = 3} \quad \text{or} \quad \mathbf{a = 5}}$$

SECTION C: Long Answer & Competency Based Questions MATHS MANTHAN | (15 Marks)

Q11. Find the ratio in which $P(4, m)$ divides the line segment joining the points $A(2, 3)$ and $B(6, -3)$. Hence find m. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Step 1: Finding the Ratio Using x-Coordinate [2.5 Marks]
Given: $A(2, 3)$, $B(6, -3)$, and $P(4, m)$. Let P divide AB internally in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

Equating the abscissa of P to 4:

$$\frac{k(6) + 1(2)}{k + 1} = 4 \implies 6k + 2 = 4(k + 1) \implies 6k + 2 = 4k + 4 \implies 2k = 2 \implies k = 1$$

Thus, the ratio is $\mathbf{1 : 1}$ (meaning P is the midpoint of AB).

Step 2: Finding m Using y-Coordinate [2.5 Marks]
Substituting $k = 1$ into the ordinate formula:

$$m = \frac{1(-3) + 1(3)}{1 + 1} = \frac{-3 + 3}{2} = \frac{0}{2} = 0$$

$$\boxed{\text{Ratio } = \mathbf{1 : 1} \qquad \text{and} \qquad \mathbf{m = 0}}$$

Q12. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Dividing into 4 Equal Parts via Successive Midpoints [1 Mark]

Let P, Q, R divide AB into 4 equal parts ($AP = PQ = QR = RB$). Then Q is the midpoint of AB, P is the midpoint of AQ, and R is the midpoint of QB.

We know the midpoint formula:

Formula Used: Midpoint $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Step 2: Finding Q (Midpoint of AB) [1.5 Marks]

$$Q = \left(\frac{-2+2}{2}, \frac{2+8}{2}\right) = \left(\frac{0}{2}, \frac{10}{2}\right) = (\mathbf{0}, \mathbf{5})$$

Step 3: Finding P (Midpoint of AQ) [1.25 Marks]

$$P = \left(\frac{-2+0}{2}, \frac{2+5}{2}\right) = \left(\mathbf{-1}, \frac{\mathbf{7}}{\mathbf{2}}\right)$$

Step 4: Finding R (Midpoint of QB) [1.25 Marks]

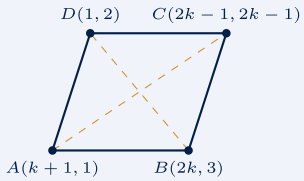
$$R = \left(\frac{0+2}{2}, \frac{5+8}{2}\right) = \left(\mathbf{1}, \frac{\mathbf{13}}{\mathbf{2}}\right)$$

Required points are $\left(\mathbf{-1}, \frac{\mathbf{7}}{\mathbf{2}}\right)$, $(\mathbf{0}, \mathbf{5})$, and $\left(\mathbf{1}, \frac{\mathbf{13}}{\mathbf{2}}\right)$

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Q13. If the points $A(k+1,1)$, $B(2k,3)$, $C(2k-1,2k-1)$ and $D(1,2)$ are the vertices of a parallelogram $ABCD$, find the value of k. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Parallelogram Diagonal Property [1 Mark]
Given: Vertices $A(k+1,1)$, $B(2k,3)$, $C(2k-1,2k-1)$, and $D(1,2)$. In parallelogram $ABCD$, diagonals AC and BD bisect each other.

We equate the midpoints of diagonals AC and BD:

Formula Used: Midpoint of AC = Midpoint of BD

Step 2: Computing Midpoints of AC and BD [2 Marks]

$$\text{Midpoint of } AC = \left(\frac{(k+1)+(2k-1)}{2}, \frac{1+(2k-1)}{2}\right) = \left(\frac{3k}{2}, k\right)$$

$$\text{Midpoint of } BD = \left(\frac{2k+1}{2}, \frac{3+2}{2}\right) = \left(\frac{2k+1}{2}, \frac{5}{2}\right)$$

Step 3: Solving Equations & Checking Mathematical Consistency [2 Marks]
Equating the abscissae (x-coordinates):

$$\frac{3k}{2} = \frac{2k+1}{2} \implies 3k = 2k+1 \implies k = \mathbf{1}$$

Equating the ordinates (y-coordinates):

$$k = \mathbf{\frac{5}{2}}$$

Consistency Note: For $k=1$, Midpoint of $AC = \left(\frac{3}{2}, 1\right)$ whereas Midpoint of $BD = \left(\frac{3}{2}, \frac{5}{2}\right)$; for $k = \frac{5}{2}$, abscissae give $\frac{15}{4} \neq 3$ (and no permutation of A, B, C, D gives matching x and y midpoints). Thus, the printed vertex coordinates contain a **data inconsistency / misprint**, so no single k satisfies all four vertices simultaneously:

$\boxed{x\text{-midpoints yield } \mathbf{k = 1}}$ whereas $\boxed{y\text{-midpoints yield } \mathbf{k = \frac{5}{2}}}$

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